

Payal Singh

Staff platform engineer working across data systems, DevOps, and AI-native systems

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I build the operating layer around PostgreSQL, shared data platforms, and model-driven workflows. My work centers on explicit contracts, reliable automation, bounded authority, and checks that determine what happens next.

PROFESSIONAL RECORD

Staff Data Engineer, Lore

June 2026-present

I lead and deliver work on managed PostgreSQL platforms, data access governance, and operational automation. Current work includes GitOps-based access controls and self-service workflows across GCP and GKE, built with Terraform and GitHub Actions, plus monitoring and failure alerting for shared data services.

Senior Database Reliability Engineer, OmniTI to Credativ to Instacluster (NetApp)

2013-June 2026

I built and operated PostgreSQL systems across reliability, data integrity, capacity, performance, audit, access control, and database migration. This work established the database and operational foundation I now apply to platform and AI-native systems.

SELECTED INDEPENDENT WORK

- Built model-driven workflows in which deterministic verification controls acceptance and explicit permission classes bound authority.
- Implemented durable state, audit trails, failure handling, and human review for long-running automated work.
- Built database migration assessment and conversion paths into PostgreSQL with deterministic stages, validation loops, structured reports, and manual review for ambiguous changes.
- Implemented data-quality gates that reject or quarantine bad input and route failures for review.

SELECTED WRITING

- [Postgres War Stories Part 1: Postgres outages that aren't Postgres bugs](#)
- [Postgres War Stories Part 2: multixact wraparound, TOAST corruption, and torn pages](#)
- [The agent that forged its own grade](#)
- [The agent is not the system. Postgres is.](#) This is an account of an implemented foundation and a redesign still in progress.

SPEAKING

- **Upcoming:** [Using PostgreSQL as a Control Plane for Reliable AI Workflows](#) - Postgres Summit US 2026, September 30, 2026, New York City
- [Postgres as an AI Control Plane: Building RAG + MCP Workflows Inside the Database](#) - SCaLE 23x, March 6, 2026 - [Organizer-hosted slides](#)
- [Securing Your Data on PostgreSQL](#) - SCaLE 16x, March 9, 2018
- [Pandas - Lazy at Mating, Lively at Mining!](#) - SCaLE 16x, March 10, 2018
- [Think Your Postgres Backups and Disaster Recovery Are Safe? Let's Talk.](#) - SCaLE 15x, March 2, 2017

EDUCATION

- **M.S. Computer Science**, University of Maryland, Baltimore County
- **B.E. Computer Science**, Panjab University